Colby Dobson

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EDUCATION

University of San Francisco, San Francisco, CA

- **M.S. in Computer Science; *AI Concentration*** — **August 2023 - May 2026**
- **B.S. in Computer Science** — **August 2021 - May 2025**
  - *Minor in Mathematics*
- **President, Club Men's Volleyball** — **August 2023 - May 2026**

WORK EXPERIENCE

Research Assistant — **June 2026 - Present**

- Developing a foundation model for astronomical spectroscopy to outperform existing redshift pipelines, enable real-time rare-object detection, improve survey targeting, and increase robustness to noise
- Co-authoring a series of papers presenting domain-specific architectural advances and rare-object discoveries
- Designing interpretable prediction heads to isolate per-source spectral features while retaining performance
- Conducting rigorous HPC experiments on millions of spectra using controlled ablations and benchmarking

University of San Francisco, Computer Science Tutor & TA — **August 2024 - December 2025**

- Diagnosed implementation and design issues while explaining complex concepts at an elementary level
- Graded Cloud Computing assignments and debugged networking and configuration issues in office hours

Sláinte Volleyball Club, Head Coach — **September 2021 - July 2025**

PROJECTS

Embedded Systems Arcade Game — **Spring 2025 - Present**

- Built an ST7796S driver on the Pico SDK for RP2040, achieving 92% of theoretical max fps without visible tearing, despite memory constrained to 2/3 a framebuffer and SPI bandwidth below one frame per refresh
- Brought up distributed Raspberry Pis with custom hardware, GPIO/UART, and real-time input arbitration
- Designed fault-tolerant timing protocols for multi-device sync with 1 controller and many players on 2 lines

Alpha-Zero Style Go Engine — **Spring 2026**

- Built a self-play pipeline with policy/value CNNs, PUCT MCTS, and player-pool sampling from prior models
- Implemented replay tooling and candidate-vs-champion evaluation gating to compare checkpoints, promote stronger models, and unify data generation, training, and evaluation in a continuous learning loop

Agentic Assistant — **Spring 2026**

- Co-architected a multi-user agent runtime as microservices supporting task execution with tool use and multi-step orchestration across planning, retrieval, and actions beyond single-call LLM interactions
- Built personalized RAG memory using vector databases and embedding models to persist user preferences, retrieve long-term context, and ground agent responses and actions in user history

RISC-V Emulator — **Spring 2023**

- Implemented a RISC-V emulator in C with instruction decoding and execution for basic instruction types, and extended its set-associative cache simulator with multiword block fills and LRU replacement
- Built a five-stage RISC-V processor in Digital with hazard tracking, forwarding, stalls, and pipeline flushing
- Designed a dual-bank memory datapath with interleaved 64-bit banks to support misaligned byte, word, and doubleword loads and stores across memory-word boundaries

Reverse Proxy — **Fall 2024**

- De-facto lead on a team of engineers in partnership with Consilient Labs working to develop a scalable and efficient reverse proxy as the primary layer of security for a prod API with fine-grained permission control

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Rust, JavaScript, Java, RISC-V Assembly

Systems: Linux, Embedded Systems, Firmware Bring-Up, Multithreading, HW/SW Integration, GPIO/UART, IPC, Real-Time Constraints, Concurrency, Distributed Systems, Networking, Computer Architecture, Protocols

Infrastructure: Git, Docker, AWS, CI/CD, Microservices, Actions, K8s, REST, SQL, noSQL, OAuth, Prometheus/Grafana

AI/ML: PyTorch, CNNs, Transformers, Agent Workflows, Agent Infrastructure, LLM Integration, RAG, Self-Play/RL

COURSEWORK

Operating Systems, Systems Foundations, Compilers, Advanced Algorithms, Machine Learning, Bayesian Deep Learning, Program Synthesis, Advanced Linear Algebra, Cloud Computing